LAB # 8 : Gamma, Exponential, and Laplace Distributions

Introduction

This lab is concerned with three important continuous probability distributions: Gamma, Exponential, and Laplace distributions. The Gamma and Exponential distributions play an important role in queuing theory and reliability analysis, while the Laplace distribution is widely used in signal processing and data science.

Gamma Distribution

The gamma distribution models the elapsed time before a fixed number of occurrences of a randomly occurring event, like calls to a pizza place or defective products in a factory. The distribution is completely specified by two parameters, α, the total number of occurrences needed, and β, the mean waiting time between events. They are known as the shape and scale parameters, respectively. The density function is given by

$$f(x) = \begin{cases} \dfrac{1}{\beta^{\alpha}\Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta}, & 0 < x < \infty,\ \alpha > 0,\ \beta > 0 \\ 0, & \text{elsewhere.} \end{cases}$$

Instead of β, the mean number of occurrences per interval is sometimes specified instead. This is known as the rate parameter, $\lambda = 1/\beta$.
There are four basic functions in R for calculating the probability for gamma distribution. In R, the gamma distribution is parameterized using the rate parameter $\lambda = \dfrac{1}{\beta}$ by default. You may also specify scale $= \beta$.

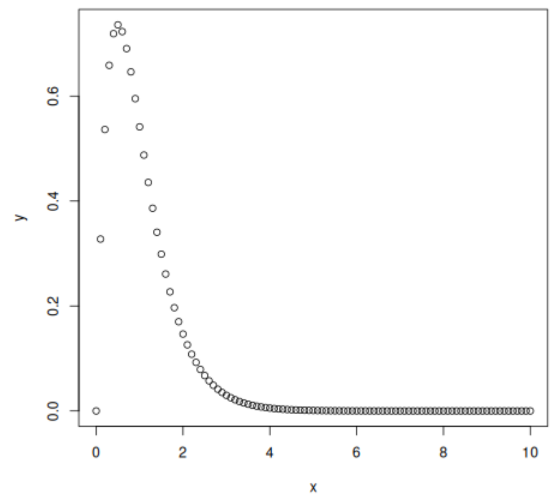
- $dgamma(x, \alpha, \lambda)$ returns probability density function of the gamma distribution with shape parameter α and rate parameter λ. This is useful in creating density plots.

```
x<-seq(0,10,by=0.1)
y<-dgamma(x,2,2)
plot(x,y)
```

- $pgamma(x, \alpha, \lambda)$ returns the cumulative probability $P(X \leq x)$ for gamma distribution.

```
pgamma(4,2,0.5)
[1] 0.5939942
pgamma(1:5,2,0.5)
[1] 0.09020401 0.26424112 0.44217460 0.59399415 0.71270250
```

- $qgamma(p, \alpha, \lambda)$ is the inverse cdf of the distribution. It returns the value x such that $pgamma(x, \alpha, \lambda) = p$.




```
qgamma(c(0.25,0.5,0.75),2,0.5)
[1] 1.922558 3.356694 5.385269
pgamma(1:6,2,0.5)
[1] 0.09020401 0.26424112 0.44217460 0.59399415 0.71270250 0.80085173
```

- $rgamma(n, \alpha, \lambda)$ generates n random values from the distribution.

```
rgamma(5,2,0.5)
[1] 3.139347 2.885717 2.891813 4.371655 8.564252
rgamma(5,2,scale=2)
```

Example 1: Calls to a customer service come at an average rate of one call per three minutes.

(a) What is the probability that more than an hour elapses before 25 calls come in?

(b) What is the 95th percentile for time needed for 5 calls to come in?

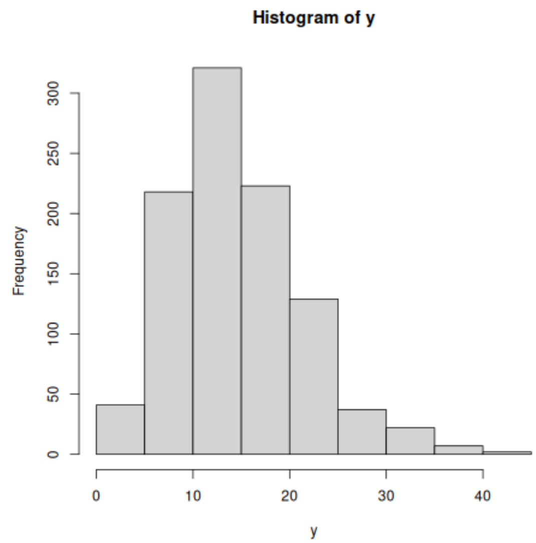
(c) Simulate waiting times for 5 calls 1000 times and produce a histogram of the results.

Sol.

```
(a)         1-pgamma(60,25,1/3)
            [1] 0.8432274
            1-pgamma(60,25, scale=3)
            [1] 0.8432274
(b)         qgamma(0.95,5,1/3)
            [1]  27.46056
(c)         y <- rgamma(1000,5,1/3)
            hist(y)
```

Exponential Distribution

This distribution models the waiting time between identical and independent randomly occurring events such as calls to a pizza place or defective products in a factory.



This is a special gamma distribution for which $\alpha = 1$ and the density function is given as

$$f(x) = \frac{1}{\beta} e^{-x/\beta}, \quad x \geq 0.$$

There are four basic functions in R similar to gamma distribution. By default, R uses mean number of occurrence $\lambda = 1/\beta$.

- $dexp(x, \lambda)$ returns the probability density function and used to create the density plots.

```
x<-seq(0,5,by=0.1)
jpeg(file="Exp.jpeg")
y<- dexp(x,2)
plot(x,y)
dev.off()
```

- $pexp(x, \lambda)$ returns the cumulative probability $P(X \leq x)$ and x can be a vector.

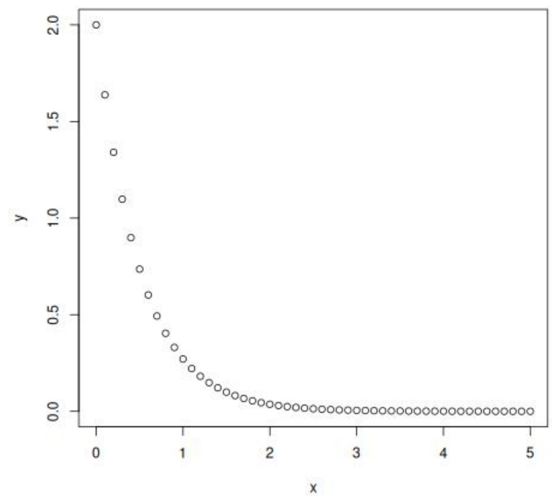
```
pexp(0.25,2)
[1]  0.3934693
pexp(c(0.25,0.50,0.75,1),2)
[1] 0.3934693 0.6321206 0.7768698 0.8646647
```

- $qexp(p, \lambda)$ is the inverse distribution function and it returns the value x such that $pexp(x, \lambda) = p$.

```
qexp(c(0.2,0.4,0.6,0.8),2)
[1] 0.1115718 0.2554128 0.4581454 0.8047190
```

- $rexp(n, \lambda)$ generates n random numbers from the exponential distribution with rate λ.

```
rexp(5,2)
[1] 0.7388777 0.6136742 1.0594232 0.1278826 1.2034926
```



Example 2: Calls to a customer service line come at an average rate of 6 every 5 minutes.

(a) What is the probability that at least 3 minutes elapse without a call?

(b) What is the 95th percentile for time between calls?

(c) Simulate waiting time between 100 pairs of calls. Plot the results.

Sol.

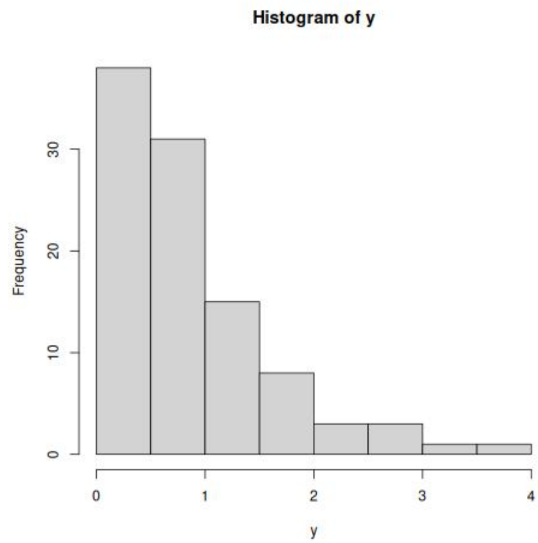
(a)
```
1-pexp(3,6/5)
[1] 0.02732372
```
(b)
```
qexp(0.95,6/5)
[1] 2.496444
```
(c)
```
y<- rexp(100,6/5)
hist(y)
```

Laplace Distribution

The Laplace distribution (also called the double exponential distribution) is a continuous probability distribution that is symmetric around its mean and has heavier tails than the normal distribution.
A random variable X is said to have a Laplace distribution with location parameter μ and scale parameter $b > 0$ if its density function is given by

$$f(x) = \frac{1}{2b} \exp\left(-\frac{|x-\mu|}{b}\right), \quad x \in \mathbb{R}.$$

We write $X \sim \text{Laplace}(\mu, b)$. The mean and median of this distribution are both equal to μ, and its variance is $2b^2$.
There are no built-in Laplace functions in base R. We define the density and distribution functions as follows:




```
# Density function
dlaplace <- function(x, mu, b) {
(1/(2*b)) * exp(-abs(x - mu)/b)
 }
# Cumulative distribution function
plaplace <- function(x, mu, b) {
ifelse(x < mu,
0.5 * exp((x - mu)/b),
1 - 0.5 * exp(-(x - mu)/b))
 }
# Random number generation  using Inverse Transform method
rlaplace <- function(n, mu, b) {
u <- runif(n, -0.5, 0.5)
mu - b * sign(u) * log(1 - 2*abs(u))
 }
```

Alternatively, functions for the Laplace distribution are available in the `VGAM` package:

- **`dlaplace(x, location=mu, scale=b)`** – density function
- **`plaplace(q, location=mu, scale=b)`** – cumulative distribution function
- **`qlaplace(p, location=mu, scale=b)`** – quantile function
- **`rlaplace(n,location=mu, scale=b)`** – simulation

Example 3: In a signal processing application, the noise X in a sensor reading follows a Laplace distribution with $\mu = 0$ and scale $b = 2$.

(a) What is the probability that the noise exceeds 3?

(b) Generate 200 random noise values and plot the histogram.

Sol.

```
# Install VGAM if not already installed, not required if using function approach
# install.packages("VGAM")
```

```
library(VGAM)
# (a) Probability P(X > 3)
1 - plaplace(3, 0, 2)
# (b) Simulation
noise <- rlaplace(200, 0, 2)
hist(noise, main = "Histogram of Laplace Noise",
               xlab = "Noise", col = "lightblue")
mean(noise)
```

Exercises:

1. Suppose that the time, in hours, required to repair a heat pump is a random variable X having a gamma distribution with parameters $\alpha = 2$ and $\beta = 1/2$. What is the probability that on the next service call

   (a) at most 1 hour will be required to repair the heat pump?

   (b) at least 2 hours will be required to repair the heat pump?

2. The length of time for one individual to be served at a cafeteria is a random variable having an exponential distribution with a mean of 4 minutes. What is the probability that a person is served in less than 3 minutes on at least 4 of the next 6 days?

3. In image compression, the difference between original and reconstructed pixel values can be modeled by a Laplace distribution with $\mu = 0$ and scale $b = 1.5$.

   (a) Find the probability that the absolute error exceeds 2.

   (b) Find the probability that the error lies between -1 and 1.

   (c) Generate 300 random errors and plot a histogram.